AIM-5014: Ai Safety & Security (3 credits)

Fall 2026

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Class Hours: Wednesday 3:15-5:15pm
Classroom: 205 Lexington Avenue 401

Assignment 3: Due Thursday Nov 5th

Assignment 3 makes use of AIGoat, a GUI-based red-teaming tool that you can spin up either locally or via colab via this notebook. The framework works best when run on a GPU so be sure to change your runtime-type to, ideally, an A100 or T4 GPU. Log in as a non-Admin (Alice, etc.) to complete all problems. Submit your completed assignment as a separate doc with, depending on the question, a completion screenshot or a one paragraph answer.

When you first interact with the chat interface, the response time will be quite slow. All subsequent queries should be extremely fast. The AIGoat GUI can be confusing. When you see "admin" in the upper-right hand corner, that does **not** mean that you are logged in as admin. Instead, it means that if you want to switch to the admin role you can click that button.

The location of the level-indicator is also a bit confusing. The change the level (0, 1, 0r 2), click the small lock icon in the upper right.

A RAG poisoning

1. **LLM09: RAG Knowledge poisoning**

   To access this question, from the homepage first click **Start the challenges**, and then click on the "LLM09: RAG Knowledge poisoning" tile.

   Note that this question first involves adding a poisoned document to the RAG datastore. When adding documents, first click the hamburger menu icon (three parallel horizontal lines), in the upper right-hand corner and then click "RAG."

   To show completion for this problem, cut-and-paste a screenshot showing the chat interface responding using the poisoned RAG database, i.e., if you added a doc that told the LLM to price all items at $.01, then the chat response should clearly indicate that all items are $.01.

B MCP Attacks

To access the MCP attack problems, starting from the home page first click the hamburger menu icon (three parallel horizontal lines) in the upper right-hand corner and then click "MCP."

To show successful completion of the MCP problems, you'll need to cut-and-paste a screen shot with AIGoat's "Evidence accepted" tab lit up.

1. **MCP01: Token Mismanagement & Secret Exposure**
2. **MCP03-1: Tool Description Poisoning**
3. **MCP02-1: Scope Creep / Capability Discovery vs Authorization**

C Agent attacks

To access the Agent attack problems, starting from the homepage first click the hamburger menu icon (three parallel horizontal lines) in the upper right-hand corner and then click "Agent."

1. **LLM03-1: Excessive Agency via Tools**

   Instructions: Set the defense level to 0 and ask the agent to refund order 1003. Read the steps list. Find the **issue_refund** step, its arguments, and its decision. Repeat the same request at Level 1, then Level 2. At Level 2 the run pauses: read the tool name and arguments in the dialog before you choose Approve or Deny. Compare the three runs. What changed between levels, and what stayed the same?

2. **ASI02-1: Tool Misuse**

   At Level 0 ask the agent to apply coupon STAFF100 to order 1003. Read the **apply_coupon** step and the impact notice that names the discounted order. At Level 1 repeat the plain request. Then add an instruction-override sentence in front and compare: input validation reads the wording, not the coupon code. At Level 2 repeat the plain request and record the control id that denies it. Then try a customer code such as WELCOME20 and confirm normal use still works. In **your own words**, describe how increase the security level changes the tool behavior.

3. **ASI06-2: Memory Scan at Level 2**

   Save the planted policy note with the example button. At Level 0 run a refund request that points to the shop notes. Confirm the note is injected and a refund is requested. This is what completes the lab. At Level 1 send the same request and note that the note is still included. At Level 2 send it again. The note stays listed with a dropped by memory.scan chip, and no refund is scored.

   To show completion, cut-and-paste a screenshot clearly showing that the injected note was dropped by the memory scan.